

Reason for Using $1.5 \times \text{IQR}$ in Outlier Detection

The value 1.5 in the formulas $Q1 - 1.5 \times \text{IQR}$ and $Q3 + 1.5 \times \text{IQR}$ is a widely accepted statistical rule used for identifying outliers. The Interquartile Range (IQR) measures the spread of the middle 50% of the data. By extending the range by 1.5 times the IQR on both sides, a reasonable boundary is established to distinguish typical observations from unusually high or low values.

The less and greater outlier ranges are calculated as follows:

Less Outlier Range = $Q1 - 1.5 \times \text{IQR}$

Greater Outlier Range = $Q3 + 1.5 \times \text{IQR}$

Observations that fall below the less range or above the greater range are considered outliers. The multiplier 1.5 is an empirical value that provides a balance between detecting genuine outliers and retaining normal observations within the dataset. As a result, the $1.5 \times \text{IQR}$ rule has become a standard method for outlier detection in statistical analysis and box plots.